Types of Blockchain

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Blockchain technology has evolved into a versatile tool with various applications across industries. Understanding the different types of blockchain is essential for selecting the right solution for specific needs. Broadly categorized into public, private, consortium, and hybrid blockchains, each type offers unique characteristics, benefits, and use cases. Public blockchains enable open access and decentralization, while private blockchains prioritize security and control. Consortium blockchains serve collaborative networks, and hybrid blockchains combine features of both public and private models. This article discusses types of blockchain in detail.

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What is Blockchain Technology?

Blockchain technology is a decentralized, distributed ledger system that securely records and verifies transactions across a network of computers. It allows multiple parties to share and access the same data without a central authority.

1. **Decentralization**: Unlike traditional databases controlled by a single entity, blockchain operates on a peer-to-peer network. This decentralization reduces the risk of data manipulation and single points of failure.
2. **Distributed Ledger**: Every participant in the network has access to a copy of the entire blockchain, ensuring that all transactions are transparent and verifiable. This distributed nature enhances trust among participants.
3. **Immutability**: Once data is recorded in a blockchain, it is nearly impossible to alter or delete. Each block is cryptographically linked to the previous one, forming a secure chain. This feature ensures data integrity and historical accuracy.
4. **Consensus Mechanisms**: Blockchain networks use various consensus algorithms (such as Proof of Work or Proof of Stake) to validate and agree on transactions. This ensures that all network participants reach a common agreement before adding new data.
5. **Smart Contracts**: Some blockchains, like Ethereum, support smart contracts. These automate processes and enforce agreements without intermediaries.

Permissionless Blockchain

A permissionless blockchain is a type of blockchain network that allows anyone to participate in the network without requiring special permissions or approvals.

1. **Open Access**: Anyone can join the network, validate transactions, and contribute to the blockchain. This openness fosters a decentralized environment where no single entity controls the network.
2. **Decentralization**: Permissionless blockchains operate on a decentralized network of nodes, which helps to distribute power and reduce the risk of censorship or manipulation by any single party.
3. **Consensus Mechanisms**: These blockchains typically use consensus algorithms such as network participants' Proof of Stake (PoS) to validate transactions and secure the network. Participants compete to solve complex mathematical problems (in the case of PoW) or stake their own tokens (in PoS) to earn the right to validate new blocks.
4. **Transparency**: All transactions on a permissionless blockchain are recorded on a public ledger, allowing anyone to view transaction history and verify data integrity.
5. **Anonymity**: While transactions are transparent, participants often remain pseudonymous. Users are identified by their public keys rather than personal information, providing a layer of privacy.

Permissioned Blockchain

A permissioned blockchain is a type of blockchain network that restricts access and participation to a select group of authorized users. Unlike permissionless blockchains, where anyone can join and validate transactions, permissioned blockchains require participants to obtain permission before they can access the network or perform certain actions.

1. **Access Control**: Only authorized participants can join the network, ensuring that all nodes are known and vetted. This allows for greater control over who can validate transactions and access data.
2. **Centralized Governance**: Typically governed by a consortium of organizations or a central authority, which makes decisions about network rules and policies.
3. **Enhanced Privacy**: Transactions and data are often more private, as sensitive information can be kept off-chain or shared only among authorized parties.
4. **Customizable Protocols**: Organizations can customize consensus mechanisms and other protocols to meet their specific needs and requirements.

Types of Blockchain

Here are the 4 types of Blockchains:

1. Public Blockchain

These blockchains are completely open to following the idea of decentralization. They don't have any restrictions, anyone having a computer and internet can participate in the

network.

1. As the name is public this blockchain is open to the public, which means it is not owned by anyone.
2. Anyone having internet and a computer with good hardware can participate in this public blockchain.
3. All the computers in the network hold the copy of other nodes or blocks present in the network
4. In this public blockchain, we can also perform verification of transactions or records

Advantages:

1. **Trustable:** There are algorithms to detect fraud. Participants need not worry about the other nodes in the network.
2. **Secure:** This blockchain is large as it is open to the public. In a large size, there is a greater distribution of records.
3. **Anonymous Nature:** It is a secure platform to make your transaction properly at the same time, you are not required to reveal your name and identity to participate.
4. **Decentralized:** There is no single platform that maintains the network, instead every user has a copy of the ledger.

Disadvantages:

1. **Processing:** The rate of the transaction process is very slow, due to its large size. Verification of each node is a very time-consuming process.
2. **Energy Consumption:** Proof of work is highly energy-consuming. It requires good computer hardware to participate in the network.
3. **Acceptance:** No central authority is there so governments are facing the issue of implementing the technology faster.

Use Cases:

Public Blockchain is secured with proof of work or proof of stake they can be used to displace traditional financial systems. The more advanced side of this blockchain is the smart contract that enabled this blockchain to support decentralization. Examples of public blockchains are Bitcoin and Ethereum.

2. Private Blockchain

can participate in the process, making it more secure than the others.

1. These are not as open as a public blockchain.
2. They are open to some authorized users only.
3. These blockchains are operated in a closed network.
4. In this few people are allowed to participate in a network within a company/organization.

Advantages:

1. **Speed:** The rate of the transaction is high, due to its small size. Verification of each node is less time-consuming.
2. **Scalability:** We can modify the scalability. The size of the network can be decided manually.
3. **Privacy:** It has increased the level of privacy for confidentiality reasons as the businesses required.
4. **Balanced:** It is more balanced as only some users have access to the transaction which improves the performance of the network.

Disadvantages:

1. **Security:** The number of nodes in this type is limited so chances of manipulation are there. These blockchains are more vulnerable.
2. **Centralized:** Trust building is one of the main disadvantages due to its central nature. Organizations can use this for malpractices.
3. **Count:** Since there are few nodes if nodes go offline the entire system of blockchain can be endangered.

Use Cases:

With proper security and maintenance, this blockchain is a great asset to secure information without exposing it to the public eye. Therefore companies use them for internal auditing, voting, and asset management. An example of private blockchains is Hyperledger, Corda.

3. Hybrid Blockchain

It is the mixed content of the private and public blockchain, where some part is controlled by some organization and other makes are made visible as a public blockchain.

1. It is a combination of both public and private blockchain.
2. Permission-based and permissionless systems are used.
3. User access information via smart contracts
4. Even if a primary entity owns a hybrid blockchain it cannot alter the transaction

Advantages:

1. **Ecosystem:** The most advantageous thing about this blockchain is its hybrid nature. It cannot be hacked as 51% of users don't have access to the network.
2. **Cost:** Transactions are cheap as only a few nodes verify the transaction. All the nodes don't carry the verification hence less computational cost.
3. **Architecture:** It is highly customizable and still maintains integrity, security, and transparency.
4. **Operations:** It can choose the participants in the blockchain and decide which transaction can be made public.

Disadvantages:

1. **Efficiency:** Not everyone is in a position to implement a hybrid Blockchain. The organization also faces some difficulty in terms of efficiency in maintenance.
2. **Transparency:** There is a possibility that someone can hide information from the user. If someone wants to get access through a hybrid blockchain it depends on the organization whether they will give or not.
3. **Ecosystem:** Due to its closed ecosystem this blockchain lacks the incentives for network participation.

Use Case:

It provides a greater solution to the healthcare industry, government, real estate, and financial companies. It provides a remedy where data is to be accessed publicly but needs to be shielded privately. Examples of Hybrid Blockchain are the Ripple network and XRP token.

4. Consortium Blockchain

It is a creative approach that solves the needs of the organization. This blockchain validates the transaction and also initiates or receives transactions.

1. Also known as Federated Blockchain.
2. This is an innovative method to solve the organization's needs.
3. Some part is public and some part is private.
4. In this type, more than one organization manages the blockchain.

Advantages:

1. **Speed:** A limited number of users make verification fast. The high speed makes this more usable for organizations.
2. **Authority:** Multiple organizations can take part and make it decentralized at every level. Decentralized authority, makes it more secure.
3. **Privacy:** The information of the checked blocks is unknown to the public view. But any member belonging to the blockchain can access it.
4. **Flexible:** There is much divergence in the flexibility of the blockchain. Since it is not a very large decision can be taken faster.

Disadvantages:

1. **Approval:** All the members approve the protocol making it less flexible. Since one or more organizations are involved there can be differences in the vision of interest.
2. **Transparency:** It can be hacked if the organization becomes corrupt. Organizations may hide information from the users.
3. **Vulnerability:** If a few nodes are getting compromised there is a greater chance of vulnerability in this blockchain

Use Cases:

It has high potential in businesses, banks, and other payment processors. Food tracking of the organizations frequently collaborates with their sectors making it a federated solution ideal for their use. Examples of consortium Blockchain are Tendermint and Multichain.

Comparative Analysis of Blockchain Types

Feature	Public Blockchain	Private Blockchain	Hybrid Blockchain	Consortium Blockchain
Access Control	Open to everyone	Restricted to specific participants	Limited to a group of organizations	Combination of public and private
Governance	Decentralized	Centralized	Semi-decentralized	Mixed governance structure
Transparency	High transparency	Low transparency	Moderate transparency	Variable transparency
Scalability	Limited scalability	High scalability	Moderate scalability	High scalability potential
Security	High due to decentralization	Lower due to centralization	Moderate security	Variable security
Transaction Speed	Slower due to consensus mechanisms	Faster transactions	Faster than public, slower than private	Variable speed
Use Cases	Cryptocurrencies, decentralized apps	Enterprise solutions, data privacy	Supply chain, banking, collaborations	Various applications need flexibility

Conclusion

In conclusion, the various types of blockchains, public, private, consortium, and hybrid, each serve distinct purposes and address specific needs. Public blockchains prioritize transparency and decentralization, while private blockchains focus on privacy and control. Consortium blockchains facilitate collaboration among multiple entities, and hybrid blockchains offer a blend of both worlds. Sidechains enable asset transfer and experimentation without affecting the main chain, whereas Layer 2 solutions enhance

scalability and speed. Understanding these types allows organizations to choose the most suitable blockchain architecture for their unique applications and goals.

Comment

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